

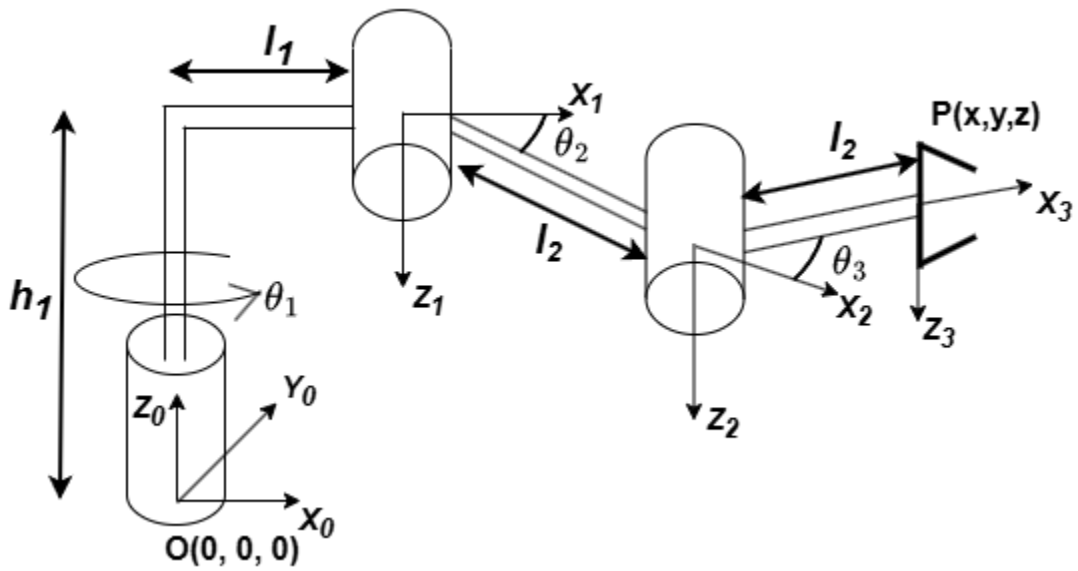
BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 461: Introduction to Robotics
Examination: Midterm Examination
Semester: Spring 2026

Pepper

Duration: 1 Hour 20 Minutes	Full Marks: 30
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1.CO1	A. Do you think robots will be essential in the near future? Justify your answer. [2] B. Nowadays, it is becoming difficult to distinguish between intelligent machines and robots. In your opinion, what features make a machine a robot? [2] C. Which industry do you think most urgently needs robots in Bangladesh? Also, do you think a family robot must be a cobot? Explain your answer. [2] D. Why do you think an intelligent robot would still obey humans even if it becomes capable of making perfect decisions independently? [2] E. Write down two applications of each of the following: [2] i. Miniature UAV (SUAV) ii. Delivery Drone	[10]
2.CO1	Team Railbot had spent months developing a maze-solving robot for a prestigious global robotics expo. The challenge followed a two-phase approach. First, the robot explored the maze to generate a complete map. And then, using that map, it had to solve the maze in the shortest possible time. Drawing from past competitions, the team equipped their robot with a high-precision laser-based sensing system for detailed environmental mapping, along with additional sensors for navigation and control. However, upon arrival, they encountered an unexpected twist. The maze was constructed entirely from reflective surfaces. The mirrors rendered their primary sensing approach unreliable, as laser readings became distorted and misleading. Faced with this limitation, the team made a rapid adjustment. They switched to an alternative sensing method capable of generating a stable 2D point cloud, allowing the robot to interpret the maze despite reflections. This strategic change proved decisive, enabling the robot to outperform competitors and secure first place. After the competition, while exploring the expo, the team encountered ALIO, an advanced industrial robotic system, equipped with precision-engineered robotic arms capable of executing highly delicate and accurate tasks. What set ALIO apart was its ability to process complex visual data streams in real time, seamlessly guiding its movements with remarkable accuracy. Its performance suggested a deeply optimized fusion of perception and graphical computation, hinting that such high-level visual intelligence could be achieved even on compact, efficient hardware platforms. A. What are the fundamental primitives of an AI-based robotic system? [1] B. Identify the two sensors used by Team Railbot. Explain the working principle of the second sensor and why the first sensor would fail while the later successfully	[10]

	mapped the maze without issues. [0.5+2.5] C. Given the conditions of the maze competition, which robotics paradigm should the maze-solving robot follow? [2] D. Which type of controller would be most suitable for ALIO, and why? [2] E. Which type of motor was used for ALIO's robotic arms? Justify your choice by comparing it with other possible options. [2]	
3.CO2	 Here, for the manipulator shown in the figure, assume the following parameters: $l_1 = 15 \text{ units}, l_2 = 30 \text{ units},$ $\theta_1 = 35^\circ, \theta_2 = 30^\circ, \theta_3 = 40^\circ$ All joint axes $Z_1, Z_2,$ and Z_3 are parallel, while Z_0 is perpendicular with them.. The manipulator operates at a constant height, $h_1 = 18 \text{ units}.$ A. Calculate the coordinates of the end effector point $P(x, y, z)$ based on the provided link lengths and joint angles. You can either use Trigonometric approach or DH parameter. [4] B. If the end effector moves from its current position by +6 units along the X-axis and -1 units along the Y-axis, while maintaining the same end-effector orientation, determine the new joint angles $\theta_1, \theta_2, \theta_3.$ [4] C. Write down the differences between Forward and Inverse Kinematics. [2]	[10]